

Lem derivadas parciales wirtinger

Lema 1. Sea $f \in \mathcal{C}^1(\Omega)$ con $\Omega \subset \mathbb{C}$ abierto, entonces

$$\frac{\partial f}{\partial x}(z) = \frac{\partial f}{\partial z}(z) + \frac{\partial f}{\partial \bar{z}}(z) \quad \wedge \quad \frac{\partial f}{\partial y}(z) = i \left(\frac{\partial f}{\partial z}(z) - \frac{\partial f}{\partial \bar{z}}(z) \right),$$

donde $\frac{\partial}{\partial z}$ y $\frac{\partial}{\partial \bar{z}}$ son las derivadas de Wirtinger.

Referenciado en

- Prop-regla-cadena-wirtinger